ENSO: Coupled Dynamics, Monitoring, and Impacts

class notes and activities

Use alongside the lecture El Niño–Southern Oscillation (ENSO).

1. ENSO basics and the mean Pacific (lecture: definitions, neutral state, thermocline)

Fill in the key distinctions.

	El Niño	**La Niña**
Central/eastern Pacific SST		
Easterly trade winds		
Eastern Pacific thermocline		
Walker circulation / convection		

Why does the neutral-state equatorial Pacific typically have a deeper thermocline in the west than in the east?

Define the Southern Oscillation (atmospheric component of ENSO).

2. Activity 1 Predict the coupled response (3 min; lecture Activity 1)

Scenario: Equatorial easterly trade winds weaken for several weeks. For each variable, predict the *sign* of the anomaly and explain one physical link.

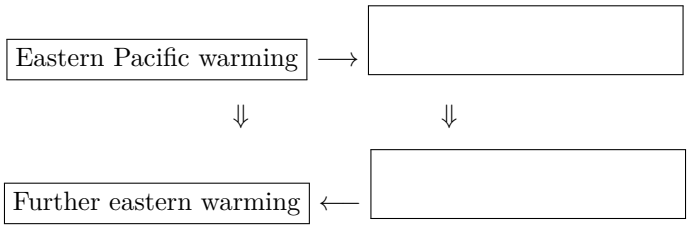
1. Eastern equatorial Pacific thermocline depth: _____. Why?

2. Eastern Pacific SST: _____. Why?

3. Central Pacific deep convection: _____. Why?

3. Bjerknes feedback (lecture: positive ocean–atmosphere feedback)

Complete the feedback chain for a developing El Niño.



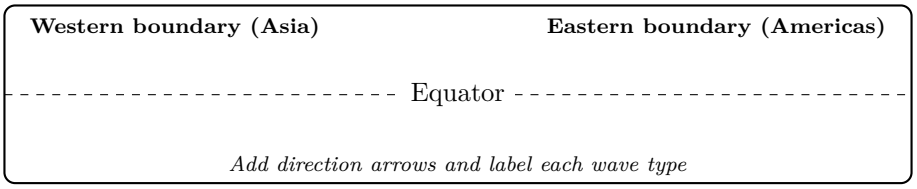
Why is this feedback *positive*? What does it fail to explain?

4. Delayed oscillator (lecture: Kelvin and Rossby waves)

Complete the mechanism table.

Signal	**Direction / location**	**Impact on eastern thermocline or ENSO**
Downwelling Kelvin wave		
Upwelling Rossby wave		
Reflected upwelling Kelvin wave		

On the schematic, add arrows showing the initial Kelvin and Rossby wave paths, then the reflected Kelvin wave. Label "warming" and "delayed cooling."

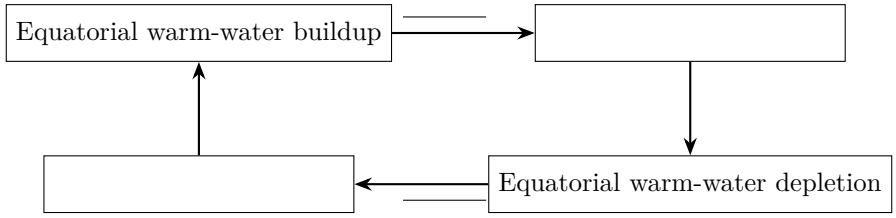


Explain in one or two sentences why this is called a *delayed* negative feedback.

5. Recharge–discharge oscillator (lecture: equatorial heat content)

Distinguish thermocline *tilt* from basin-mean equatorial thermocline *depth*.

Complete the conceptual cycle using the terms: recharge, discharge, El Niño, La Niña, and equatorial heat content.



During an established El Niño, why can equatorial upper-ocean heat content be falling even while eastern Pacific SST is still warm?

6. Activity 2 Why does El Niño end? (3 min; lecture Activity 2)

Scenario: Eastern Pacific SSTs remain warm, but equatorial warm-water volume has fallen below its climatological value.

1. Use the *recharge oscillator* to predict the likely subsequent SST tendency.

2. Use the *delayed oscillator* to identify an oceanic-wave route to the same outcome.

One similarity and one difference between the two oscillator theories:

7. ENSO monitoring (lecture: Niño regions, ONI, RONI, SOI, ocean observations)

Record what each index or observation measures and one limitation.

Measure	What it measures	Why useful / caveat
Niño 3.4 / ONI		
RONI		
SOI		
Warm-water volume / thermocline		

8. Activity 3 Diagnose an event (3 min; lecture Activity 3)

Hypothetical observations: Niño-3.4 SST anomaly +1.2°C; SOI negative; easterly trades weaker than normal; eastern thermocline deeper than normal.

1. Which ENSO phase is suggested? Which two indicators support your diagnosis?

2. Why is it important to examine both atmospheric and oceanic variables?

3. Can one month's SST anomaly establish a *sustained* ENSO event? Explain.

9. Global impacts and synthesis (lecture: teleconnections, monsoons, tropical cyclones)

Trace the physical pathway from tropical SST anomalies to a remote climate response.

Tropical SST anomaly → [ ] → [ ]

→ Regional rainfall / storm-track change

How does El Niño tend to affect Atlantic hurricane activity? Name the dynamical mediator, not just the sign.

Reflection: Why are regional ENSO impacts probabilistic rather than deterministic?